ST 538 Group 4 Project 2

Introduction

Vehicle loan default prediction presents a distinct set of challenges relative to other retail credit products. Unlike mortgage or personal loan portfolios, two-wheeler financing in emerging markets such as India serves a large population of first-time borrowers with little or no bureau history, making traditional score-based underwriting insufficient on its own. This analysis uses the CRIF High Mark vehicle loan dataset to develop a feature set capable of identifying first-EMI default risk across both scored and thin-file borrower segments.

Data and Processing

The dataset comprises 233,154 disbursed loans with a binary default indicator capturing payment failure on the first EMI due date. A notable characteristic of this population is that 56% of borrowers carry no scoreable bureau history, which necessitates treating thin-file and scored segments with care rather than imputing or ignoring the distinction. Among scored borrowers, the CRIF CNS score serves as the single strongest individual predictor and is encoded here as a 20-level ordered factor preserving the full risk ranking across bands.

Feature engineering focuses on translating domain knowledge into model-ready variables: loan structure risk (LTV, implied down payment), borrower demographics (age at disbursement, employment type), identity verification depth (KYC document count), credit utilisation and delinquency history across primary and secondary accounts, EMI burden relative to existing obligations, and recent bureau activity including new account uptake and hard enquiries.

A composite stress score aggregating six binary risk flags is also constructed as a diagnostic and interpretability aid. Initial cross-tabulation confirms that the stress score tracks default rate monotonically, rising from 15% at score 0 to 49% at score 5, providing early evidence that the engineered features carry meaningful signal.

```
# load dependencies
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.4.3

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'stringr' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(lubridate)
library(stringr)
library(knitr)
```

```
Warning: package 'knitr' was built under R version 4.4.3
```

There are no missingness in the data! Pretty sweet!

Feature engineering

```r
# validation
cat("Rows:", nrow(train_dat), "\n")
```

```
Rows: 233154
```

```r
cat("Total columns after engineering:", ncol(train_dat), "\n")
```

```
Total columns after engineering: 73
```

```r
cat("Net new columns:", ncol(train_dat) - 41, "\n\n")
```

```
Net new columns: 32
```

```r
cat("--- LTV high flag (>80%) ---\n")
```

```
--- LTV high flag (>80%) ---
```

```r
print(table(train_dat$ltv_high_flag, useNA = "ifany"))
```

```

     0      1
151266  81888
```

```r
cat("\n--- Age segment ---\n")
```

```

--- Age segment ---
```

```r
print(table(train_dat$age_segment, useNA = "ifany"))
```

```

       prime senior_risky  young_risky
      218475          867        13812
```

```r
cat("\n--- Employment type ---\n")
```

```

--- Employment type ---
```

```r
print(table(train_dat$EMPLOYMENT_TYPE, useNA = "ifany"))
```

```

     Salaried Self employed          <NA>
        97858        127635          7661
```

```r
cat("\n--- CNS score factor levels (confirm all levels matched) ---\n")
```

```
--- CNS score factor levels (confirm all levels matched) ---
```

```r
print(levels(train_dat$cns_score_factor))
```

```
 [1] "No Bureau History Available"
 [2] "Not Scored: More than 50 active Accounts found"
 [3] "Not Scored: No Activity seen on the customer (Inactive)"
 [4] "Not Scored: No Updates available in last 36 months"
 [5] "Not Scored: Not Enough Info available on the customer"
 [6] "Not Scored: Only a Guarantor"
 [7] "Not Scored: Sufficient History Not Available"
 [8] "M-Very High Risk"
 [9] "L-Very High Risk"
[10] "K-High Risk"
[11] "J-High Risk"
[12] "I-Medium Risk"
[13] "H-Medium Risk"
[14] "G-Low Risk"
[15] "F-Low Risk"
[16] "E-Low Risk"
[17] "D-Very Low Risk"
[18] "C-Very Low Risk"
[19] "B-Very Low Risk"
[20] "A-Very Low Risk"
```

```r
cat("Unmatched descriptions (NAs in factor):",
    sum(is.na(train_dat$cns_score_factor)), "\n")
```

```
Unmatched descriptions (NAs in factor): 0
```

```r
cat("\n--- Thin-file borrowers ---\n")
```

```
--- Thin-file borrowers ---
```

```r
print(table(train_dat$thin_file_flag, useNA = "ifany"))
```

```
     0      1 
103369 129785 
```

```r
cat("\n--- Primary any overdue ---\n")
```

```
--- Primary any overdue ---
```

```r
print(table(train_dat$pri_any_overdue_flag, useNA = "ifany"))
```

```
     0      1 
206879  26275 
```

```r
cat("\n--- Recent delinquency flag ---\n")
```

```
--- Recent delinquency flag ---
```

```r
print(table(train_dat$recent_delinquency_flag, useNA = "ifany"))
```

```
     0      1 
214959  18195 
```

```r
cat("\n--- Stress score distribution ---\n")
```

```
--- Stress score distribution ---
```

```r
print(table(train_dat$stress_score, useNA = "ifany"))
```

```
     0      1      2      3      4      5 
 31573  44945 110556  45181    813     86 
```

```r
cat("\n--- avg_acct_age_months (first 10 rows) ---\n")
```

```
--- avg_acct_age_months (first 10 rows) ---
```

```r
print(data.frame(
  raw    = head(train_dat$AVERAGE_ACCT_AGE, 10),
  parsed = head(train_dat$avg_acct_age_months, 10)
))
```

```
           raw parsed
1    0yrs 0mon      0
2   1yrs 11mon     23
3    0yrs 0mon      0
4    0yrs 8mon      8
5    0yrs 0mon      0
6    1yrs 9mon     21
7    0yrs 0mon      0
8    0yrs 2mon      2
9    4yrs 8mon     56
10   1yrs 7mon     19
```

```r
cat("\n--- credit_history_months (first 10 rows) ---\n")
```

```
--- credit_history_months (first 10 rows) ---
```

```r
print(data.frame(
  raw    = head(train_dat$CREDIT_HISTORY_LENGTH, 10),
  parsed = head(train_dat$credit_history_months, 10)
))
```

```
           raw parsed
1    0yrs 0mon      0
2   1yrs 11mon     23
3    0yrs 0mon      0
4    1yrs 3mon     15
5    0yrs 0mon      0
6    2yrs 0mon     24
```

```
7   0yrs 0mon      0
8   0yrs 2mon      2
9   4yrs 8mon     56
10  1yrs 7mon     19
```

```r
# default rate by stress score table
stress_table <- train_dat %>%
  group_by(stress_score) %>%
  summarise(
    n            = n(),
    n_default    = sum(LOAN_DEFAULT, na.rm = TRUE),
    .groups      = "drop"
  ) %>%
  mutate(
    n_no_default  = n - n_default,
    default_rate  = paste0(round(n_default / n * 100, 1), "%")
  ) %>%
  select(
    `Stress score`  = stress_score,
    `Total (n)`     = n,
    `No default`    = n_no_default,
    `Default`       = n_default,
    `Default rate`  = default_rate
  )

knitr::kable(
  stress_table,
  format      = "simple",
  caption     = "Default Rate by Stress Score",
  align       = c("c", "r", "r", "r", "r"),
  format.args = list(big.mark = ",")
)
```

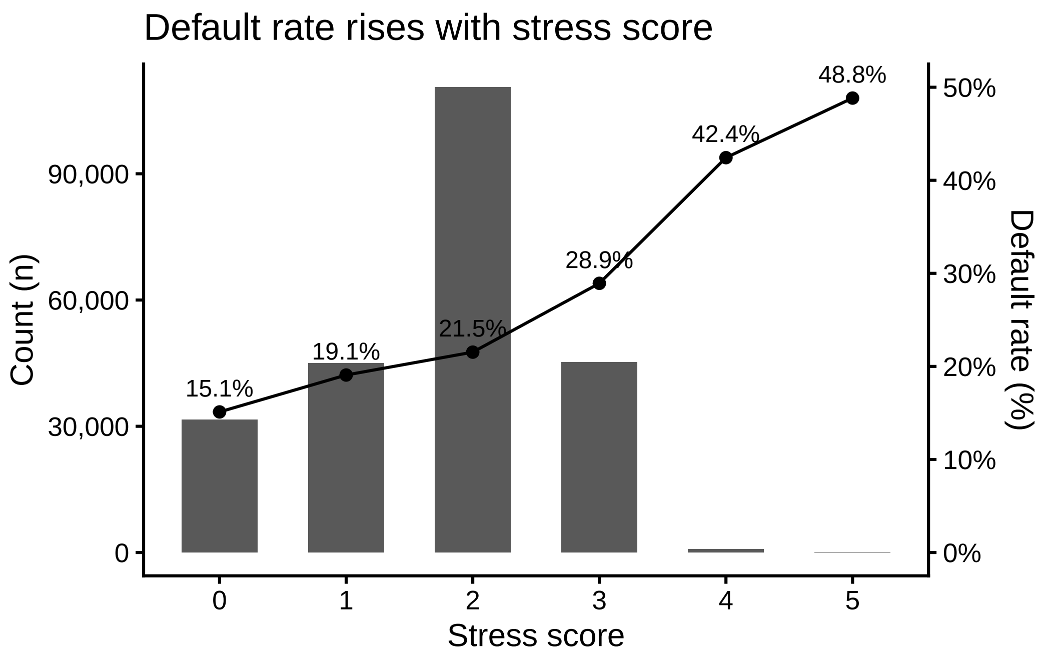
Table 1: Default Rate by Stress Score

Stress score	Total (n)	No default	Default	Default rate
0	31,573	26,802	4,771	15.1%
1	44,945	36,371	8,574	19.1%
2	110,556	86,748	23,808	21.5%
3	45,181	32,110	13,071	28.9%
4	813	468	345	42.4%

Stress score	Total (n)	No default	Default	Default rate
5	86	44	42	48.8%

```r
# default rate by stress score graphically
stress_summary <- train_dat %>%
  group_by(stress_score) %>%
  summarise(
    n             = n(),
    n_default     = sum(LOAN_DEFAULT, na.rm = TRUE),
    .groups       = "drop"
  ) %>%
  mutate(default_rate = n_default / n * 100)

scale_factor <- max(stress_summary$n) / 50

ggplot(stress_summary, aes(x = factor(stress_score))) +

  geom_col(aes(y = n), width = 0.6) +

  geom_line(aes(y = default_rate * scale_factor, group = 1)) +
  geom_point(aes(y = default_rate * scale_factor)) +

  geom_text(
    aes(y = default_rate * scale_factor, label = paste0(round(default_rate, 1), "%")),
    vjust = -0.9, size = 3.2
  ) +

  scale_y_continuous(
    name     = "Count (n)",
    labels   = scales::comma,
    sec.axis = sec_axis(~ . / scale_factor, name = "Default rate (%)",
                        labels = scales::label_number(suffix = "%"))
  ) +

  theme_classic(base_size = 12) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor   = element_blank()) +

  labs(
    title = "Default rate rises with stress score",
    x     = "Stress score"
```

```
)
```



Summary of EDA and feature engineering: Thirty-five percent of loans are high loan to value (LTV) of >80%, combined with a large portion of borrowers having thin credit history (%), it shows a meaningful concentration of risk. The vast majority (94%) of borrowers fall in the prime age band, 13,812 young borrowers (<22 years of age) are worth watching. Fifty-six percent of borrowers are “thin file” meaning they have <12 month of credit history. This makes sense, in India about half of first time car buyers are using lending for the first time. Thin and normal are two different populations, we may want to split the dataset and create a model for thin credit and one for normal, or focus our efforts on normal only–the model will likely suffer trying to work on both.

Stress score is working well as a risk signal. Default raises from 15% at score 0 to ~49% at score 5 (3x). Most of the population sits at 2 (21% default), probably thin-file + high-LTV, but no active delinquncy.The jump from 2 to 3 is steep (21-29%), pri_any_overdue_flag and/or recent_delinquncy_flag are likely doing a lot of work. Scores 4 and 5 are small segments with high default rates (42-49%), it’s really up to us if we want to keep them, probably should. Score 0 is not zero (15.1%), even the cleanest borrowers carry default risk. We can use this stress_score as a standalone feature, but it’s worth noting that components might actually capture more non-linear interactions than the score itself.

```
# engineered feature index table
feature_index <- tribble(
  ~Category,          ~Feature,
```

```r
  "Loan terms",       "ltv_high_flag",
  "Loan terms",       "down_payment_amt",
  "Loan terms",       "down_payment_pct",
  "Demographics",     "age_at_disbursal",
  "Demographics",     "age_segment",
  "Demographics",     "disbursal_month",
  "Demographics",     "disbursal_quarter",
  "Demographics",     "disbursal_year",
  "Demographics",     "fy_end_flag",
  "KYC",              "kyc_doc_count",
  "KYC",              "strong_kyc_flag",
  "Bureau score",     "cns_score_factor",
  "Bureau score",     "thin_file_flag",
  "Bureau score",     "thin_file_reason",
  "Bureau score",     "cns_score_clean",
  "Primary credit",   "pri_active_util_ratio",
  "Primary credit",   "pri_any_overdue_flag",
  "Primary credit",   "pri_leverage_ratio",
  "Primary credit",   "pri_avg_loan_size",
  "Primary credit",   "pri_sanction_disbursed_gap",
  "Secondary credit", "sec_any_overdue_flag",
  "Secondary credit", "sec_has_exposure_flag",
  "EMI / DSCR",       "total_emi_burden",
  "EMI / DSCR",       "new_loan_emi_ratio",
  "Recent activity",  "recent_delinquency_flag",
  "Recent activity",  "inquiry_to_new_acct_ratio",
  "Recent activity",  "avg_acct_age_months",
  "Recent activity",  "credit_history_months",
  "Recent activity",  "short_history_flag",
  "Composite",        "stress_score",
  "Composite",        "high_stress_flag"
)

# collapse each category's features into a single comma-separated string
feature_index_wide <- feature_index %>%
  group_by(Category) %>%
  summarise(Features = paste(Feature, collapse = ", "), .groups = "drop")

knitr::kable(feature_index_wide, format = "simple", col.names = c("Category", "Features"),
             caption = "Feature Index Table")
```

Table 2: Feature Index Table

Category	Features
Bureau score	cns_score_factor, thin_file_flag, thin_file_reason, cns_score_clean
Composite	stress_score, high_stress_flag
Demographics	age_at_disbursal, age_segment, disbursal_month, disbursal_quarter, disbursal_year, fy_c
EMI / DSCR	total_emi_burden, new_loan_emi_ratio
KYC	kyc_doc_count, strong_kyc_flag
Loan terms	ltv_high_flag, down_payment_amt, down_payment_pct
Primary credit	pri_active_util_ratio, pri_any_overdue_flag, pri_leverage_ratio, pri_avg_loan_size, pri_
Recent activity	recent_delinquency_flag, inquiry_to_new_acct_ratio, avg_acct_age_months, credit_hist
Secondary credit	sec_any_overdue_flag, sec_has_exposure_flag